

Multiple Choice (10 marks)

1. Let `x = 8`. What is `x >> 1` in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8
2. Let `x = 1`. What is `x << 3` in Python 3?
 - (a) 1
 - (b) 3
 - (c) 8
 - (d) 16
3. Which is a valid expression in Python 3.5?
 - (a) `sort('ab')`
 - (b) `sorted('ab')`
 - (c) `"ab".sort()`
 - (d) `1/0`
4. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. Are Python variable names case-sensitive?
 - (a) Yes
 - (b) No
 - (c) It's machine-dependent
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the `strip()` function removes all leading and trailing whitespace from a string.
7. True or False: The output of `'abc'[::-1]` in Python 3 is `'abc'`, indicating that the string remains unchanged.
8. True or False: An Internet Protocol (IP) address is assigned to a device when it connects to the Internet.
9. True or False: $O(n^2)$ represents the largest growth rate among the functions $O(1)$, $O(n)$, $O(n^2)$, and $O(\log n)$.
10. True or False: The value of the Python expression `1 + 3 % 3` is 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the selection sort algorithm, focusing on how the number of comparisons remains constant regardless of the initial arrangement of elements. Why is this significant in sorting?
12. Discuss how creating the negative of a digital image through RGB value transformation can be considered a lossless process. Explain the significance of this transformation in image manipulation.

End of Paper